SUPERCENTER PROCESS ANALYSIS AND STAFFING MODEL DESIGN

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Abstract

UPS (United Parcel Service) Global Freight Forwarding specializes in transporting large shipments of freight globally and domestically. They operate four supercenters across the country that handle paperwork for these shipments. These supercenters have handled 575,793 orders of global freight in the past 4 months since UPS shifted to a supercenter model, and each shipment is documented by agents without a set process flow, with limited efficiency tracking. UPS speculates that the supercenters are overstaffed, resulting in excessive use of budget and increased headcount. For the solution, the team conducted time studies at 3 supercenter locations in order to conduct a detailed process analysis of the agents' desk operations. In addition, the team compiled the time studies and its relevant metrics into a staffing calculator, which adjusts the headcount needed based on the volume of freight forecasted on a monthly basis. The proposed calculator and process analysis decreases labor downtime and improves budget and capacity utilization. Based on the calculator, the team determined that 2 of the 3 supercenters are understaffed, though this finding comes with limitations discussed in the report. The team recommends that the staffing calculator be used monthly in order to allow for a more stable headcount and to meet shipment demand accurately. Additionally, a final economic analysis has been conducted to qualify the findings made by the proposed solution

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Key Words: time study, standardized workflow, resource allocation, economic analysis, process analysis

Acknowledgements

We would like to sincerely thank the UPS Global Freight Forwarding team for their partnership and support throughout this project. We are especially grateful to our company sponsors for opening their supercenter operations to us, facilitating our time studies across the three locations, and providing the context needed to understand the agent desk workflows. Their transparency about staffing challenges and willingness to engage with our findings made it possible to develop a solution that is both practical and operationally grounded.

We extend our appreciation to our faculty advisor Dr. Karthik Chandrasekaran for his guidance at every stage of this project. His feedback pushed us to sharpen our methodology, approach our time study data with greater analytical discipline, and ensure that our staffing calculator rested on a sound and defensible foundation. His mentorship was instrumental in helping us translate field observations into a structured and meaningful set of recommendations.

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1. Introduction

1.1 Background

United Parcel Service is one of the largest logistics companies in the world providing transportation, distribution, trade, and brokerage services in over 220 countries across the globe. UPS was founded in 1907 in Seattle, Washington as a small postage delivery service. Since then, the company has grown to over 490,000 employees worldwide and transports over 23.3 million packages each day. Currently, UPS is headquartered in Atlanta Georgia. UPS Global Freight Forwarding is a division of UPS that specializes in transportation of bulk amounts of ocean, air and ground freight. This is different from the usual package shipping, which deals with lighter packages and parcels. UPS Global Freight Forwarding takes pride in their reputation of never saying no to any cargo, and often spearheads into uncharted markets such as transportation of medical supplies.

UPS Global Freight Forwarding has facilities all across the country and are divided into 4 different types. Most of them are small dispatch or operations facilities, while the larger ones are gateway facilities (depicted by rectangles in Figure 1) and supercenters (depicted by stars in Figure 1). Gateway facilities focus primarily on air freight and are located near airports LAX (Los Angeles), DFW (Dallas), ORD (Chicago), and MIA (Miami) to provide worldwide air freight services to all different regions. Supercenters serve as the network's hubs, and are designed to handle large shipment quantities while streamlining data processing for the surrounding smaller service centers.

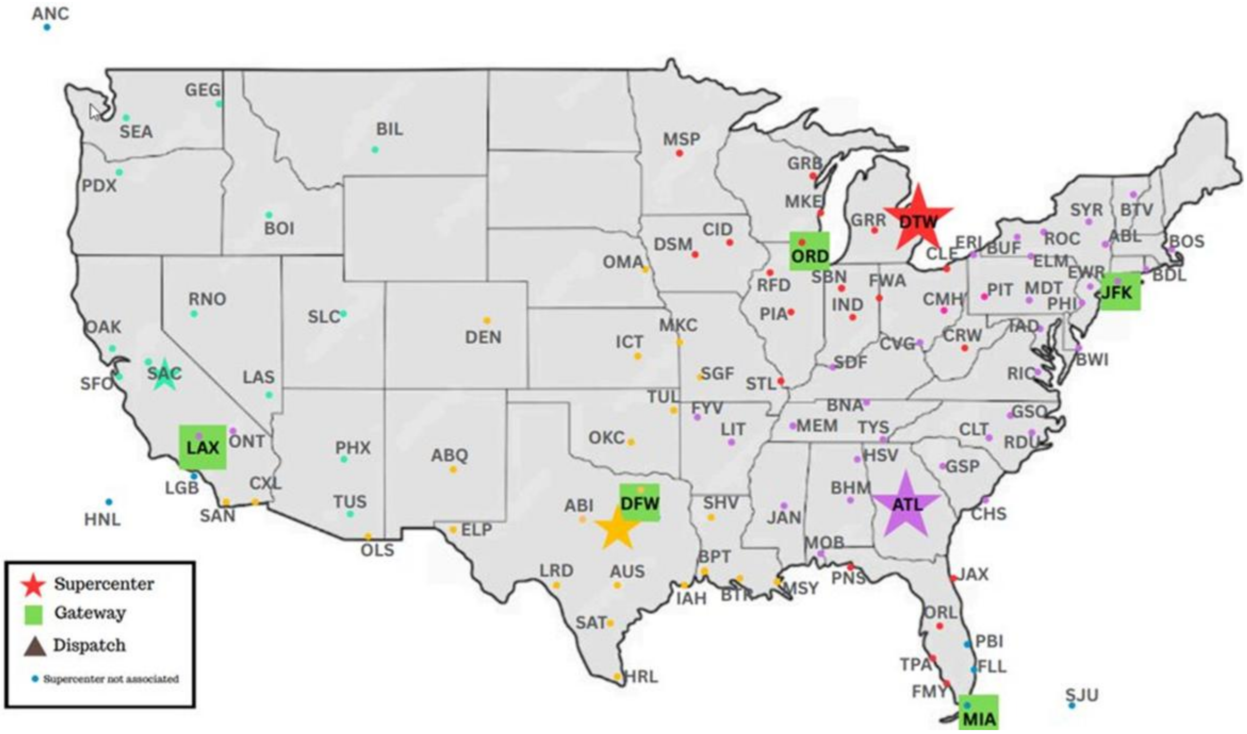

Figure 1: UPS Facility Layout

These supercenters have offices with operators who perform tasks like documentation of shipments, pick up requests (PRQ's), quote generation and various other data entry tasks. A typical set of tasks for these operators starts with the customer requesting an item to be sent over email, followed by the operator inputting the necessary data into E2K (common data entry software used in the logistics industry), after which the operator generates a quote. The customer then chooses to accept or reject the quote, and depending on the outcome, a pickup is scheduled. The final step of the shipment process is data entry, in which the agent enters the information from the BOL into the E2K system and uploads the UPS BOL/HAWB into a shared database. The agents are also in charge of reaching out to customers to address any issues or inconsistencies when they come up. Overall, while the agents' responsibilities are clearly defined, the absence of a standardized process for carrying them out leads to inconsistencies and inefficiencies in overall supercenter operations.

United Parcel Service currently lacks a standardized workflow for its supercenter operators and does not track how long daily tasks take to complete. As a result, the company does not have the necessary data to support accurate staffing decisions, increasing the risk of understaffing or overstaffing. Another issue that these supercenters are facing is, all four locations seemingly having different or varying workflows and responsibilities for agents despite working the exact same position.

These issues have led the management team to view the current operations of the supercenter as inefficient. UPS wants the University of Illinois team to conduct structured time studies to record day to day operations of these agents and to create a staffing calculator that will determine the number of agents needed depending on the volume of shipments being processed. The staffing calculator will use data from the time studies combined with demand data from each facility to give an accurate number of agents needed to process all shipments on a monthly basis.

1.2 Economic Motivation and Impact

The goal of the project is to enhance the efficiency of the UPS Global Freight Forwarding supercenter model by matching the number of staff to the actual work processed, and standardizing existing workflows. The four supercenters processed 575,793 freight orders in the last four months, handled by desk agents using the E2K system. However, the inability to track the workflow makes it difficult to evaluate the efficiency of the supercenters and match the number of staff to the actual work processed.

Staffing for the supercenters, on average, costs $3,217,084 annually per supercenter, with the agents working at $29 per hour. The cost savings that can be achieved by improving the efficiency of the supercenters can be significant, as there are four supercenters in the UPS Global Freight Forwarding organization. The time studies determine the duration of these tasks, the idle time, and the process variation by comparing multiple agents and supercenters.

Another important part of the proposed solution is the creation of a dynamic staffing calculator. The calculator will integrate time study information with shipment order volume forecasts to determine how many agents would be needed each month. Since it will be built in Excel, it does not impose any additional operational cost on UPS.

The overall intended impact is better labor utilization, reduced operational inefficiencies, and standardized workflows across the supercenter network.

2. Problem Statement and Deliverables

Problem Statement:

UPS Global Freight Forwarding lacks standardized process flow within its supercenters, resulting in an average annual staffing cost of $3.22 million per supercenter. The team is tasked with conducting time studies, developing process maps, analyzing the current state of staffing and designing a dynamic staffing calculator in Excel to determine an optimal staffing logic.

Deliverables:

To address these challenges, the project will have the following key deliverables:

1. **Time Study Data:** A comprehensive dataset compiled from virtual monitoring of desk operations, detailed and precise timestamps of shipment processing procedures and administrative tasks across all four supercenters.
2. **Standardized Process Maps:** Visual workflow maps that outline the optimal process flow for operations performed by supercenter agents. These maps will identify and eliminate repetitive process steps and ensure consistency across different facilities.
3. **Dynamic Staffing Calculator:** An Excel-based tool that integrates time study data with seasonal shipment volumes. This calculator will allow management to determine the precise headcount required on a monthly basis to meet demand without overstaffing.
4. **Economic Impact and Validation:** A final report to quantify the reduction in downtime, and budget savings achieved through the implementation of the new staffing calculator. Post-launch validation to ensure compliance with standardized process flows.

3. Team Objectives

The following objectives are deemed to be necessary for successful project completion.

1. **Initial Economic Analysis:** The preliminary budget planning for UPS supercenter staffing, the labor cost per hour per agent, and the shipments per hour data for each UPS supercenter must be observed and taken into consideration when determining the overall starting economic costs of labor.
2. **Time Study:** Time studies will be performed on desk operations across four supercenters. Each agent tasked with shipment procedures through virtual monitoring will be timed and tasks broken down into basic steps.
3. **Process Mapping:** A process map detailing a comprehensive step by step overview of the desk operations that are involved in receiving and sending a shipment order will be created. Areas of downtime, value/non value added tasks, and lean techniques will determine the necessity of each desk operation.
4. **Final Economic Analysis:** Previous and future demand in shipments will be taken into consideration, as well as number of agents per supercenter needed and hours worked per supercenter, which will facilitate the creation of a staffing calculator. This information will be determined based on the results of the time studies and Process Analysis.
5. **Staffing Calculator:** The staffing calculator in Excel will use the results of the Final economic analysis as well as previous/anticipated shipment demand in order to determine the number of agents needed per supercenter to handle shipment orders on a monthly basis.

6. **Develop Conclusions:** A summary of the findings obtained from the staffing calculator will be provided alongside other considerations.
7. **Recommendations:** A list of recommendations and course of action that satisfy the problems listed in the problem statement will be provided.

4. Time Study Analysis

4.1. Time Studies

The team coordinated with UPS leadership in order to conduct time studies virtually at three UPS supercenters: Dallas, Detroit and Atlanta. The team has determined that conducting time studies on the different desk operators will help us create a standardized process flow and assumptions for standard process times. With the data from these time studies, the team can build a dynamic staffing calculator in Excel. This calculator will help the company staff their supercenters accurately, based on their forecasted shipments per month.

From a quantitative standpoint, the team has received data on shipment volumes across each supercenter division on a weekly basis, as well as an average of $29 hourly wages for the agents. These inputs are critical for developing the staffing calculator and conducting an economic analysis, as they directly connect workload demand with labor costs and overall staffing requirements. After discussing with the company, the team finally aligned on three primary deliverables that will define the project’s success: comprehensive time studies, a standardized workflow, and a dynamic staffing calculator.

During the first two weeks, the team scheduled a meeting with one of the supervisors from the Dallas supercenter and obtained a high-level overview of the tasks that are most commonly performed by the supercenter desk agents. The supervisor's insight on the different processes allowed the team to familiarize themselves with commonly used applications and their interfaces, as well as how tasks flow from one into the other. Some of the common tasks that are performed by the desk agents include quote requests, PRQs (pick up requests), and shipment entries.

The quote request process starts with the customer sending an email to one of the agents (or to an email distribution group that later gets sent to an agent). The email typically includes key shipment details such as pickup and delivery locations, weight and dimensions, and any hazardous materials information. The agent then takes this information and puts it into the E2K system, which performs the calculations and generates the quote as a pdf. The agent then takes the quote and sends it back to the customer to await their response. The supervisor noted that some common issues found in the process include customers not providing all the necessary details to generate the quote, as well as delayed response times after receiving the quotes.

If the quote has been approved by the customer, the next step is to create a pickup request. The operator takes the information the customer previously provided and inputs it into the pick-up request section of E2K. The system takes in the information and sends it over to dispatch to get it scheduled for a pickup. Once the pickup is processed, the agent must log the details in the centralized shipment list to ensure visibility and ease of tracking for the entire team.

Shortly after the supervisor meeting the team created a spreadsheet for recording critical information collected from the time studies, as shown below in Figure 2.

Start Time	End Time	Total Time	System Used	Notes
12:20	12:22		E2K + Outlook	Responding to a request (ensuring on time delivery)
12:22	12:33		E2K - Shipment List and Shipment Entry	scheduling shipment for Angel Cellular
12:33	12:38		SSRS Sql Server	responding to request saying shipment already in MIA
12:38	12:45		Outlook	gving a shipment status update
12:45	12:47		E2K csv customer service caller menu	giving a shipment status update for EVEXIAS
12:47	12:50		Outlook	Status Update for Kimball Solutions

Figure 2: Initial Time Study Spreadsheet

The columns of the spreadsheet include the start and end times, total time, system used and notes. After conducting the first time study, the team was able to gather sufficient data on the different process times as well as expand some of the common processes that the agents do. Beyond their core duties, agents are frequently responsible for responding to status inquiries, retrieving shipment data from SQL databases, and distributing bills of lading. Not only did the first time study allow the team to truly understand the supercenter agents’ responsibilities, it also allowed the team to improve upon the spreadsheet for tracking. The first change that was made to the spreadsheet was changing from recording start and end times, to using stopwatch times for more precision. The start and end time method allows you to subtract the end time by the start time and get the total time however after discussing with the company contacts, the team concluded that precision down to the nearest second was important for the outcome of the project. The stopwatch method allows for more precise second time measurements for each process that is done. A column was created after the time studies were completed to categorize the notes into process buckets that would help during the process analysis stage. Figure 3 below depicts the improved spreadsheet that was used for the following time studies.

#	Stopwatch	Bucket	System Used	Activity Description
1	0:02:05	Shipment Entry	Teams + E2K	Inputting PRQ numbers from Teams sheet into shipment entry page
2	0:00:35	Shipment Entry	E2K	Inputting information about the order into customer list
3	0:00:20	Invoice & Data Preparation	PDF	Looking at customer's commercial invoice
4	0:09:10	Shipment Entry	E2K	Updating piece summary details; description, dimensions, package type, etc.
5	0:01:35	Invoice & Data Preparation	E2K + PDF + Calculator	Looking at customer shipping info (invoice?) and calculating total value
6	0:02:25	Invoice & Data Preparation	Excel + PDF	Inputting shipping info (part monetary values) into an excel sheet
7	0:00:45	Idle / Break / Waiting	Calculator	Waiting for calculator's currency page to load
8	0:02:00	Invoice & Data Preparation	PDF	Adding the total prices from the invoice sheets
9	0:00	Shipment Entry	E2K	Inputting information to the shipment entry page from invoice
10	0:03:35	Invoice & Data Preparation	Calculator + PDF + Excel	Adding item weights up from the invoice and inputting them to excel

Figure 3: Finalized time study Spreadsheet

After completing the other scheduled time studies with the agents from the Dallas supercenter, the team also noticed some inconsistencies between some of the agents and their work days. One of the agents was commonly observed doing tasks that the other agents (who have worked a less amount of time) did not. On top of the standard processes, this agent was also responsible for inputting shipment delays into E2K, as well as scanning and uploading paperwork. The team also learned from an agent, as well as from observation that there is significantly less idle time, and agents tend to be much busier earlier in the week (Monday through Wednesday), and the days tend to slow down as you get closer to the weekend.

The time studies at Detroit and Atlanta supercenters were conducted similarly to the Dallas one. The Atlanta supercenter was where the team encountered the most difficulties in terms of agents being rescheduled constantly, or working different shifts than expected. During each of the time studies the team uses Zoom's screen sharing and recording feature to allow for live observation of on-screen activity as shift recordings for later reference.

Following the completion of the time studies, the team sorted each task into process buckets which differentiated the type of task being performed (i.e. shipment entry vs PRQ), and the total times were tabulated. The team then created a process analysis spreadsheet, which will be discussed in more detail in Section 5.

4.2. Staffing Logic

UPS requested the team to create a staffing calculator using the data gathered from the time studies and process analysis. Initially, the definition of a staffing calculator was discussed with the primary stakeholders from UPS. The requested calculator will generate the necessary staffing for incoming demand. In the context of the supercenter, optimal staffing describes a staffing level that satisfies the incoming demand at a minimum cost while also satisfying additional constraints such as labor laws, UPS staffing guidelines, request processing delay tolerance.

From the provided SPH document, an initial staffing requirement calculation was performed.

$$Number\ of\ agents = \frac{Incoming\ Demand}{Shipments\ per\ Hour * Hours\ worked\ in\ one\ shift * Days\ worked\ in\ a\ week}$$

(1)

This model provides the minimum number of required agents to satisfy the work hours necessary to process the incoming demand. Incoming demand is the total volume of shipments to be completed, shipments per hour is the number of shipments an agent can complete in an hour, hours worked in one shift and days worked in a week represent the amount of hours worked by an individual agent in a week. This initial model was programmed on Excel and used as a prototype to demonstrate to UPS the potential input and output of the final product.

The model in Equation (1) is viable. However, it does not account for processing delay tolerance, process variability, and possible other factors that necessitate a more complex model. The team then considered a Mixed Integer Programming model. This model would also accept a demand level as an input and return the staff quantity to satisfy that demand. The constraints would represent labor laws such as working hour limits, inventory balance, delay tolerance, maximum budget, demand satisfaction, integer number of workers, minimum and maximum shift lengths, facility capacity, nonnegativity, and possibly efficiency factors. The objective function would minimize the product of total hours and hourly rate.

In the final communication on the staffing calculator between UPS and the team, UPS stated that their delay tolerance was no more than a few hours. They also stated that the group could assume UPS will have accurate quote request demand estimates, and could consider those metrics during the design and testing. Based on these considerations the team decided to avoid the Integer Programming approach and use time studies to figure out realistic productivity levels. Using this data and the logic from Equation (1), the team decided to build a simplified calculator in Excel.

5. Process Analysis

After the time studies were done, the task level observations from all three supercenters were analyzed and classified. Each task was categorized according to the type of activity involved into six different process buckets: shipment entry, shipment tracking, pick-up request, quote/rate request, communication, and idle. With this organization of tasks, the team could combine the task duration and frequency of occurrence from various agents and supercenters in order to calculate the volume-weighted productivity per category. As each agent had a different concentration of tasks, the sum of total observation time across all agents was divided by the sum of total occurrences across all agents to find the productivity metric.

The four processes for which the team is calculating productivity metrics for, are shipment entry, pickup requests, quote/rate requests, and shipment tracking. Three of these process buckets correspond directly to the staffing calculator's three primary demand inputs: quote requests, pickup requests, and data entry. Accurate demand figures were unavailable for shipment tracking processes. Shipment tracking along with the remaining categories, communication and idle, will be accounted for as a percentage of unavailable time in the calculator.

The time study data enabled the team to build a standardized process flow for each of the three primary workflows. The quote request process begins when an agent receives a customer email requesting shipment pricing, typically containing origin and destination locations along with cargo weight. The agent opens E2K, enters the rate details, reviews the system's output, and generates a quote, which is converted to a PDF invoice and returned to the customer via Outlook. A common source of delay in this process is incomplete customer information, which requires the agent to follow up before the quote can be generated.

The pickup request process is initiated once a customer accepts a quote. The agent retrieves the prior shipment details, opens the pickup entry section of E2K, inputs all required logistics information, and sends a confirmation email. The request is then dispatched to field operations with scheduling.

The shipment entry process is the most time-intensive of the three, requiring the agent to retrieve a Bill of Lading (BOL) via Teams or Outlook, input all customer and shipment freight details, shipment dimensions, and export documentation into E2K, save the BOL through ImageIndexUI, update the centralized Teams shipment list, and mark the task complete. Another observation was that this process is notably faster when shipment data is available in a CSV format that can be imported directly, reducing manual entry time by approximately half compared to manual transcription from a PDF or email.

Outside of the three primary process categories, shipment tracking was identified as the process category to which agents allocated the most significant amount of their time. This process is typically initiated when a customer, or a coworker contacts an agent, via e-mail or telephone, to request a status update on a specific freight order. The requested information can include the location of a shipment, the order status of a shipment, the delay cause of a shipment, etc. Upon receiving the request, the agent finds the shipment in E2K through a unique shipment ID such as the HAWB number, or through the customer shipment list. The agent then identifies the requested information through the E2K system. This process can include viewing updated photographs of the shipment and cross referencing information such as current location to the scheduled shipment plan.

5.1 Process Maps

Each agent's observed work can be categorized into a standardized process map including all of the agent's different responsibilities using swim lane diagrams. Within the diagram, the different lanes represent different roles within the entire process, and includes customers, inbound channel, agent, E2K system as well as documentation. Each node within the lanes represents a certain task or subtask that is being performed. The node modules include entry/exit (oval), decision (diamond), process (rectangle), subprocess (divided rectangle) and data, which are all represented in the following legend as shown in Figure 4.

Figure 4: Process Map Legend

The master process map as shown in Figure 5 visualizes every aspect of the desk operator's role and focuses primarily on how each responsibility flows from one to another in the scope of the entire operation.

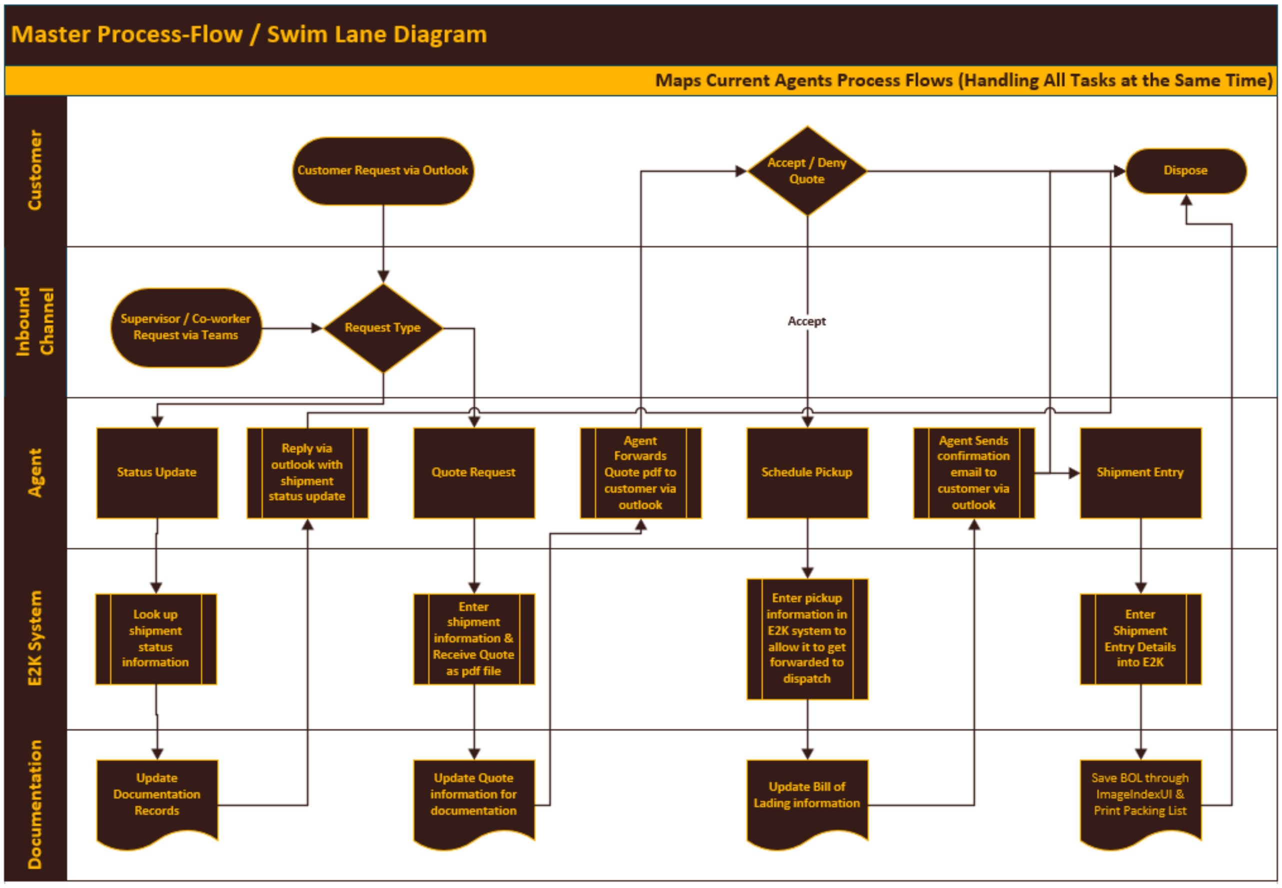

Figure 5: Master Process Map

The quote request process shown in Figure 6 takes in a request from the customer or inbound channel and first sends it to a decision module, where the agent checks if the quote request has all necessary information. If the request does not have all the necessary information to process the request, it gets redirected to the customer to complete the information. The agent moves forwards and begins the quote generation process in E2K. The data gets documented and the agents send it back to the customer for approval. If the customer denies the quote the process gets disposed of.

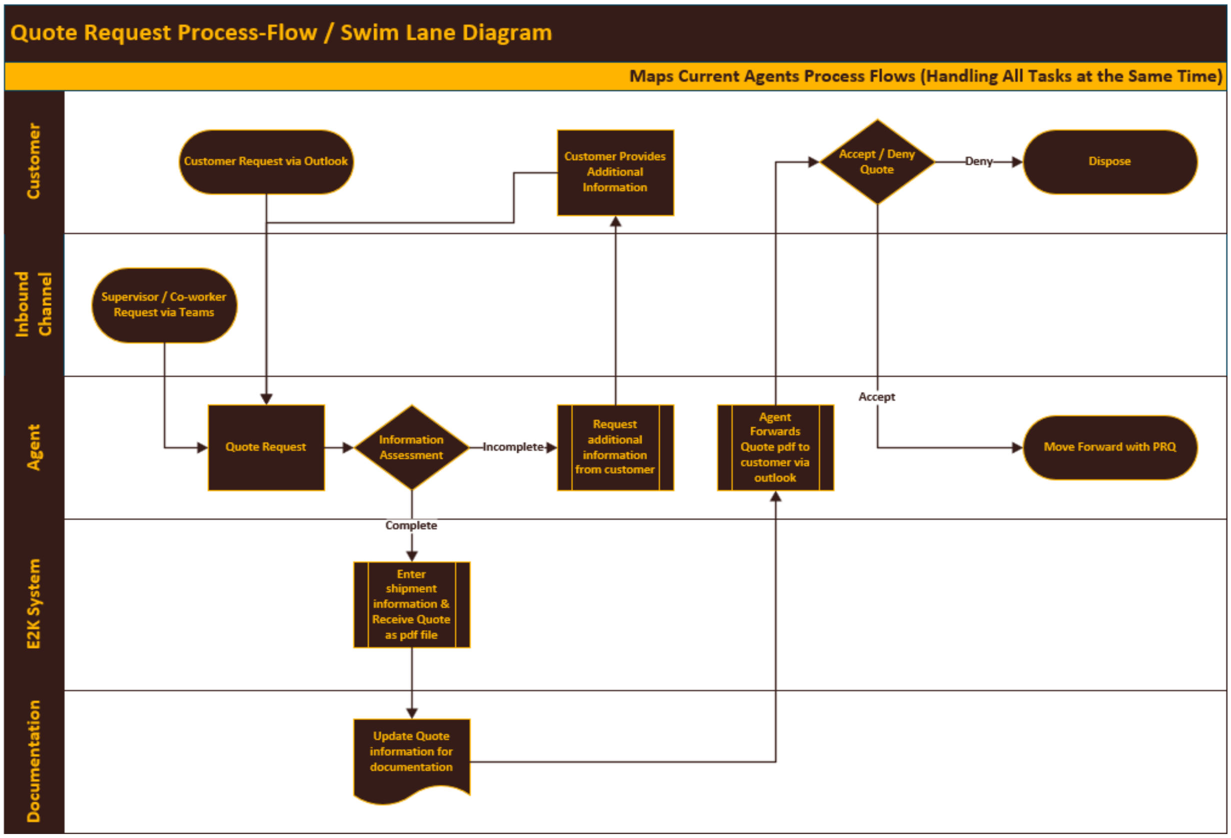

Figure 6: Quote Generation Process Map

The PRQ process in Figure 7 takes the accepted quote from the customer or the inbound channel, where the agent processes it and enters the pick up request information into the E2K PRQ system. Once all the information is entered into E2K, the operator documents and updates all the Bill of Lading Information, then forwards it, along with a confirmation email back to the customer so they are made aware of a successful PRQ.

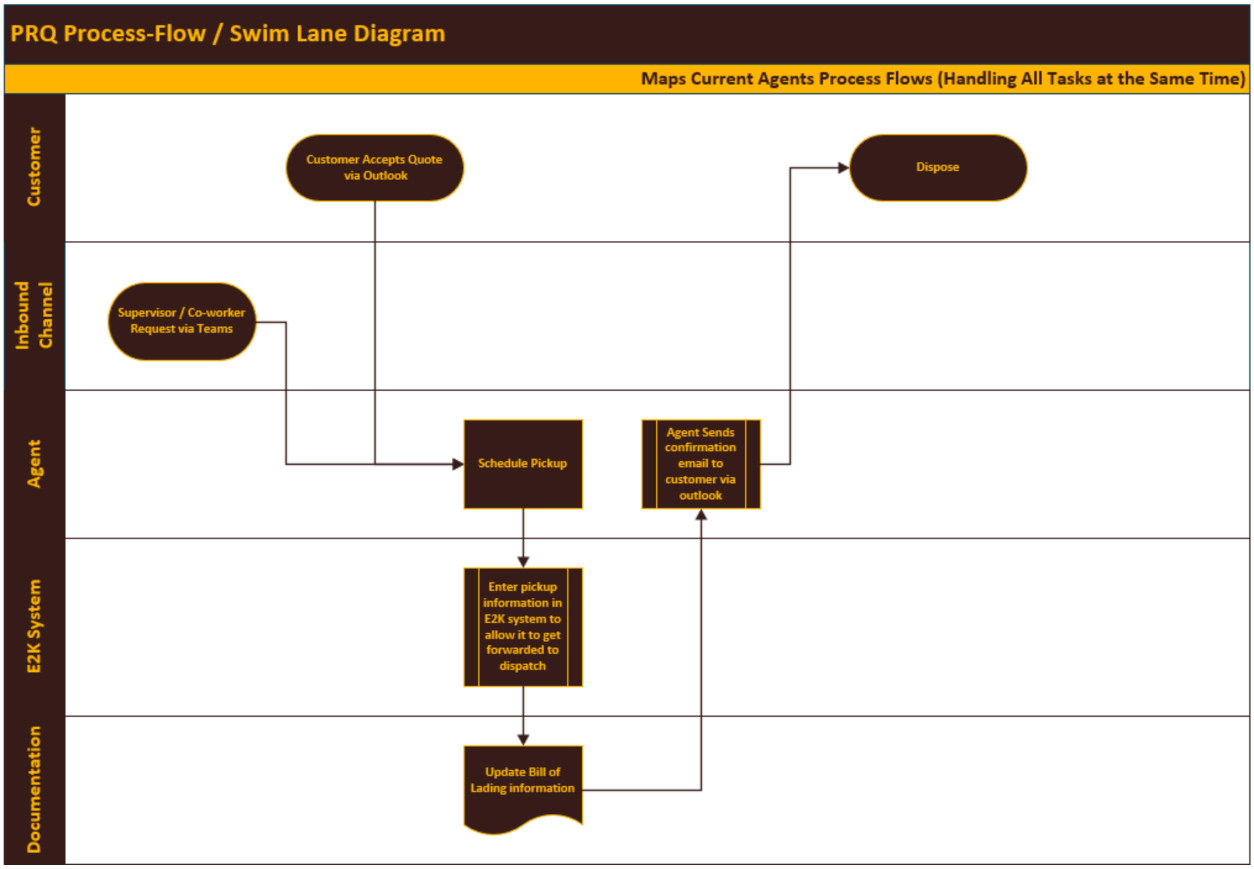

Figure 7: Pick Up Request Process Map

The shipment entry process in Figure 8 takes in the Bill of Lading from the customer or another inbound channel through teams or outlook.. If a CSV is available, all details can be imported into the E2K shipment entry system much faster. If the CSV is not available, the agent manually transcribes all of the details into the E2K system. After this step is complete, the BOL is saved through imagenIndexUI and the packing list is printed. The agent then validates the details, after which the process is complete.

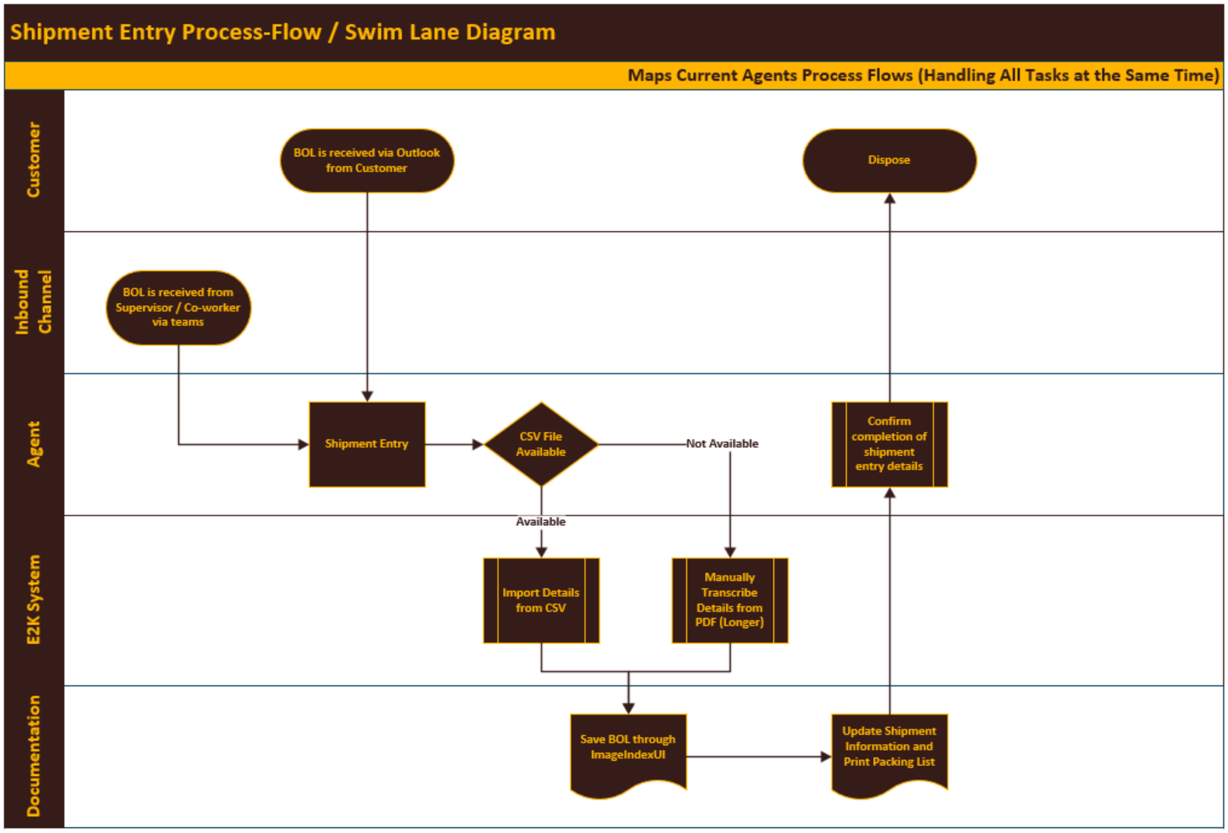

Figure 8: Shipment Entry Process Map

5.2 Findings

After categorization, for each process bucket, the team listed a set of sub-tasks that the agents would perform. Here is an example of a set of sub-tasks to complete one pickup request: receiving requests through outlook/teams, entering details into e2k, scheduling pickup requests via e2k and forwarding confirmation. The team organized these processes and sub-tasks in a spreadsheet. Figure 9 shows an example of one agent at the Detroit supercenter.

Process	Pick up Request	Email & Communications	Shipment Tracking	Idle
	Recieve requests through outlook	Check outlook inbox	Look up shipment via shipment # in e2k	idle
	Entery Details into e2k	Respond to inquires	Forward Status Overview through outlook	
	Schedule Pickup Request via e2k	Reply to status updates	Update shipment status in e2k	
	Forward confirmation	Verify customer informion		
		Scroll & Sort through inbox		
Total Time	3 hours 33 minutes	54 min	21 min	51 min
Occurences	34	19	4	15
Productivity	6.26m	2.84m	5.25m	

Figure 9: Spreadsheet Organization of Total Time, Occurrences and Productivity for an Agent

Once observation was complete, the total time spent on all instances of a given process was summed and divided by the number of times that process was completed end-to-end during the session, yielding an average minutes-per-occurrence figure referred to as the productivity metric. With this information for each process, agent, and supercenter, the team was able to conduct further analysis comparing the productivity of various agents and supercenters; as well as using them as parameters in our calculator.

Dallas emerged as the highest-performing supercenter across most process categories. Pickup requests averaged 3.88 minutes per occurrence, the lowest of any site, while quote requests averaged 5.5 minutes. Shipment entry averaged 7.9 minutes, and shipment tracking came in at 5.28 minutes. These results suggest Dallas staff have developed efficient, repeatable workflows for high-frequency tasks, particularly pickup and coordination processes. Notably, however, Dallas also recorded the highest combined communication and idle time at approximately 608 total minutes across observed sessions (286 minutes in communication, 321 minutes idle), raising a concern that overall time utilization may not be as efficient as per-task metrics alone suggest. A significant portion of capacity appears absorbed by email management, Teams coordination, and unproductive waiting periods rather than billable processing work.

Detroit showed moderate productivity across core processes, with pickup requests averaging 5.59 minutes and quote requests at 4.39 minutes. Shipment entry averaged 9.17 minutes per occurrence, and shipment tracking came in at 5.35 minutes, both slightly above Dallas. Detroit's idle time was comparatively lower at approximately 96 minutes total, and communication time was 225 minutes, suggesting a better balance between active processing and downtime than Dallas. One area of concern is shipment entry; Detroit agents logged up to 10.16 minutes per shipment entry in one observed session, indicating inconsistency in execution speed across staff members. This variation was likely driven by lack of E2K familiarity and variation across agent workflows.

Atlanta recorded the longest average cycle times across all four process categories: 8.45 minutes for pickup requests, 9.33 minutes for quote requests, 9.92 minutes for shipment entry, and 5.62 minutes for tracking. Combined idle time was approximately 129 minutes with 138 minutes in communications. Although Atlanta's agents showed a more balanced distribution of time between communication and idle periods compared to Dallas, the higher per-task times suggest underlying process inefficiencies that are likely systemic rather than individual.One observed session flagged an unusually high 20.82-minute average for pickup requests across only six occurrences, suggesting that the agent was probably in the process of training and kept referring back to their supervisor for help. Figure 10 shows a summary of this data.

City	Pickup Request	Quote Request	Shipment Entry	Shipment Tracking
Dallas	3.88m	5.5m	7.9m	5.28m
Detroit	5.59m	4.39m	9.17m	5.35m
Atlanta	8.45m	9.33m	9.92m	5.62m

Figure 10: Summary of Productivity Metrics for Each Process per Supercenter

Several inferences can be made from the aggregate data. First, shipment entry is the most time-intensive process at every location, driven by multi-system workflows requiring staff to navigate E2K, Teams, Outlook, and ImageIndexUI sequentially. Second, communication overhead is disproportionately high across all supercenters, particularly Dallas, suggesting that email and Teams management is interrupting productive work time. Instead of agents spending time scrolling on Microsoft Team or Outlook, implementing structured queue management or dedicated communication windows could meaningfully improve productivity. Visual representation of the data is shown in Figures 11, 12, 13, 14, 15.

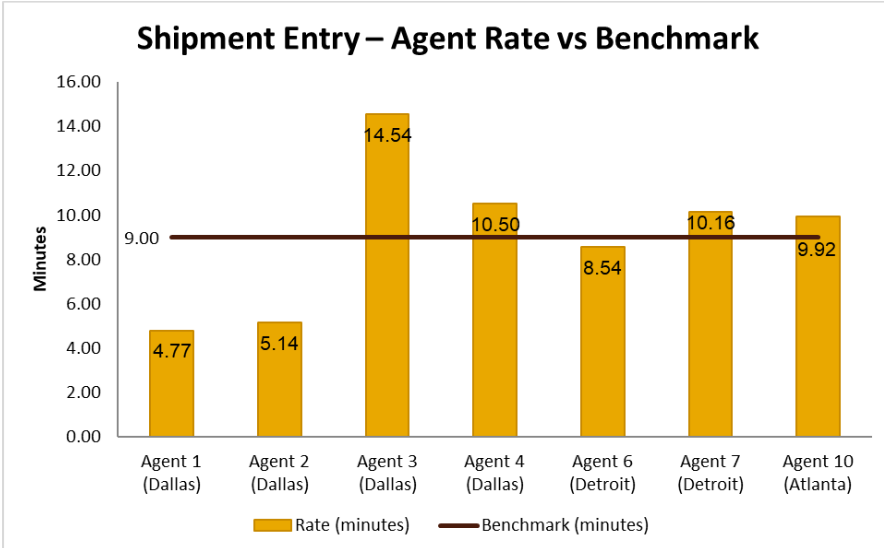

Figure 11: Shipment entry processing time by agent

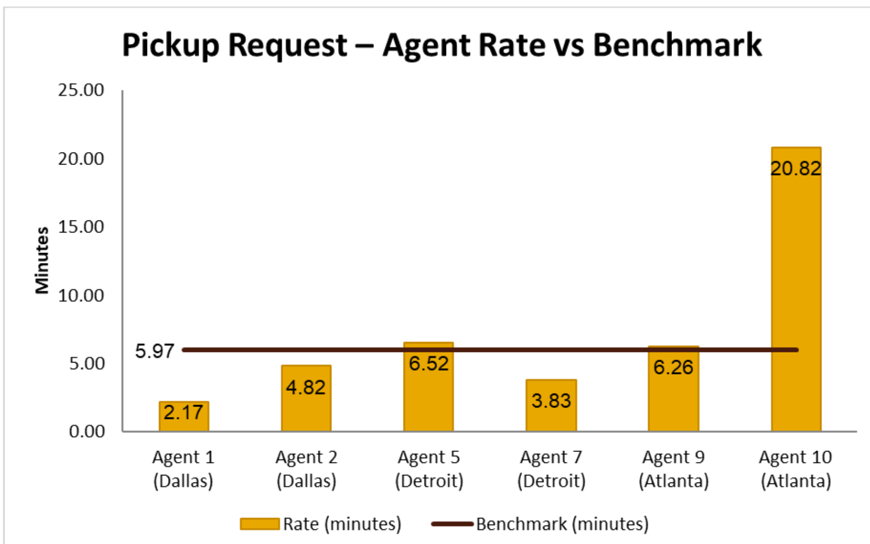

Figure 12: Pickup request processing time by agent

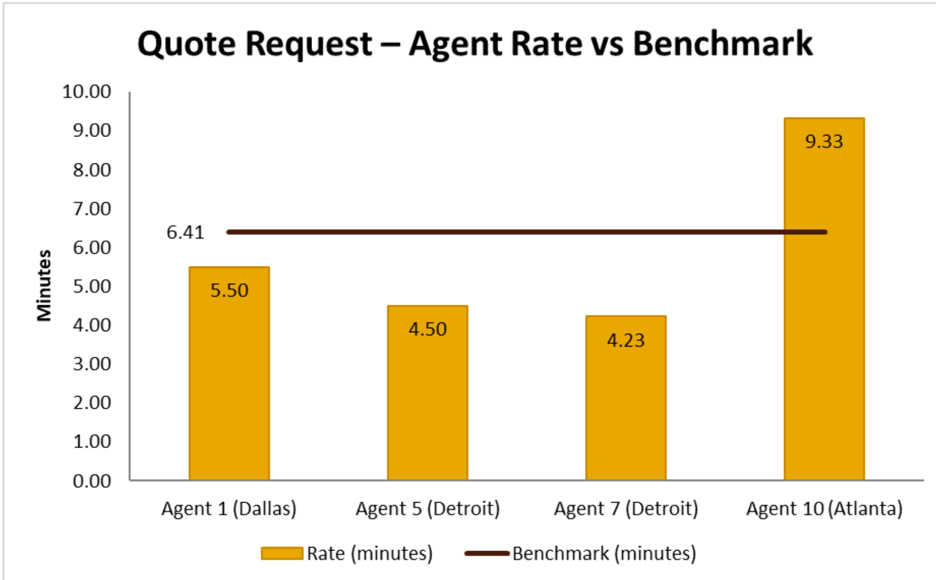

Figure 13: Quote/rate request processing time by agent

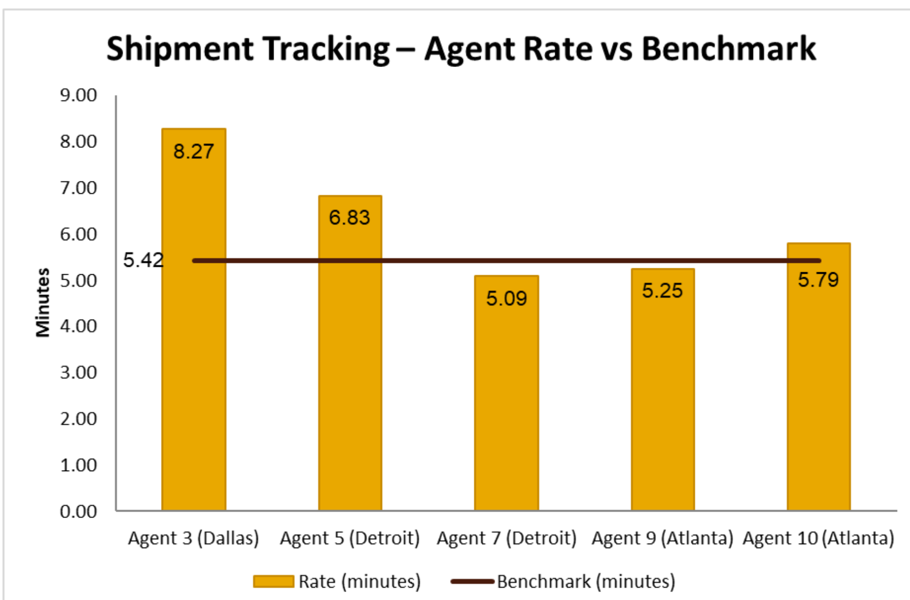

Figure 14: Shipment Tracking processing time by agent

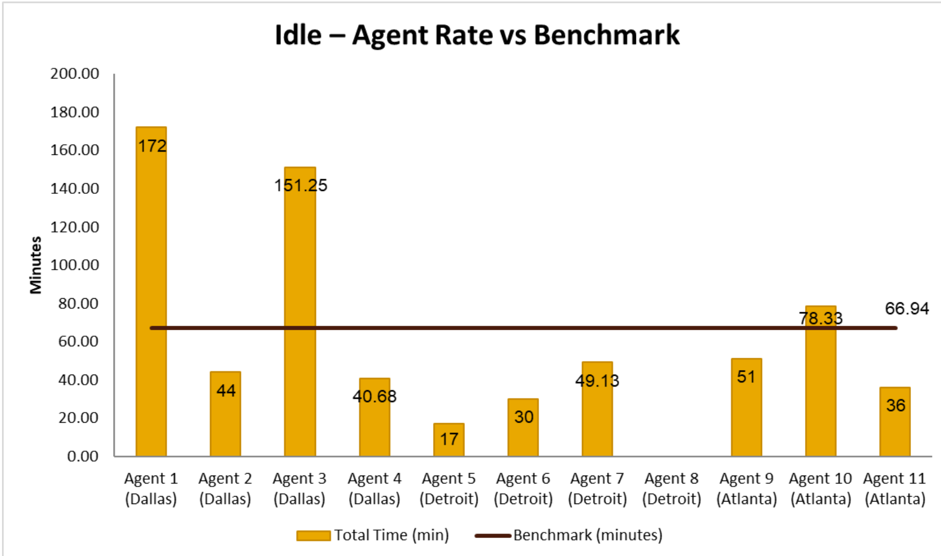

Figure 15: Idle time by agent

The team observed 6 out of 11 agents engage in shipment tracking tasks. The variation in responsibilities was caused by an unofficial separation of tasks among agents. There were 88 total occurrences of shipment tracking processes, each taking ~5.36 minutes on average. 12.08% of total observed time was spent engaging in such tasks, taking up 471.7 minutes out of 3906.2 minutes.

The process analysis also revealed workflow inconsistencies across supercenter locations that contribute to the productivity variation. Different agents perform the same role with variations in scope and responsibilities. The absence of a documented standard operating procedure means that task assignment is determined informally by supervisors and varies by day of the week, agent tenure, and division demand. The process analysis helps establish a baseline for what a standardized agent workflow should include across all three primary process buckets and serve as the foundation for productivity rate inputs used in the staffing calculator.

6. Staffing Calculator

One of the deliverables for the team is to design and develop a staffing calculator in Excel to determine the necessary agent headcount across three core operational processes: quote requests, pickup requests, and data entry. Currently, staffing decisions for the supercenters are not differentiated by process and not based on a structured, data-driven approach. The designed calculator translates input demand data into agent headcount by combining time study results with shift scheduling information and idle time inputs.

6.1 Staffing Calculation Logic

The final approach used in the formula is shown below.

$$agent\ Count_{Process} = \frac{Workload_{Process} \times Processing\ Time_{Process}}{(Days\ Worked\ per\ agent) \times (Hours\ worked\ per\ day\ per\ agent) \times (Util.Rate)} \tag{2}$$

In the above formula, workload represents the total number of incoming tasks per process, processing time per process represents the average completion time, the utilization rate represents the percentage of the day that is allocated to productive tasks, excluding non-tracked processes and idle time.

This calculation is done on a per-process basis for the quote request, pickup request, and data entry process categories. The processing time metrics, as well as idle time suggestions, are parameters decided by the team's time studies. Workload, days worked, and idle factor are user inputs into the calculator.

Although the team gathered process time data for the shipment tracking process category from the time studies, there was no available demand data. So, the calculator includes the necessary time for this process as an additional percentage of idle time, increasing the allocated unavailable time from the baseline 15%.

6.2 Data Sources

Tariff Outbound (OB) Volume Data

The primary demand input is the Tariff OB volume. This field represents the number of outbound shipments passing through the ATL service center in a given week. The data is broken out by IAF (International Air Freight) and NAAF (Non-Air and Air Freight) and is recorded on a week-ending basis as shown in Figure 16. The calculator assumes that each PRQ and data entry process leads to a completed shipment, so the forecast shipment volume for a timespan is assumed to be the PRQ and data entry demand for the same timespan.

ATL — Tariff OB Volume (Raw Data)					
W/E Date	Location	SVC Code	Tariff OB Volume	Month	Subproduct
2026-01-31	ATL	LAX	190	January	IAF
2026-01-31	ATL	LAX	162	January	NAAF
2026-02-07	ATL	LAX	188	February	IAF
2026-02-07	ATL	LAX	138	February	NAAF
2026-02-14	ATL	LAX	167	February	IAF
2026-02-14	ATL	LAX	164	February	NAAF
2026-02-21	ATL	LAX	159	February	IAF

Figure 16: ATL Outbound (OB) volume data

Email Volume Data

While pickup requests and data entry use total OB volume directly as their demand input, quote requests require an adjustment. Not every outbound shipment generates a customer inquiry. Some customers proceed directly to booking without requesting a quote. Sometimes the quote requests are not successful, but need to be accounted for as it contributes to the workload for the agent.

To tackle this, the team received a spreadsheet containing LAX email volume data as shown in Figure 17. This spreadsheet includes all the emails that the LAX division receives on a daily basis. The spreadsheet contains the ticket number associated with an email thread. For example, if a customer sends an email inquiring about the quote, a ticket number is assigned to that e-mail thread, and all further e-mails regarding that shipment will be attached to that thread and unique ticket number.

The number of unique e-mail ticket numbers for a time period was accepted as the incoming quote request demand for that period. While the team did not have access to the contents of different e-mails, the amount of total e-mails was confirmed to be directly proportional to the quote request demand when tested with LAX e-mail and staffing data.

Title	Email
CA-0326-02148	Re: [EXTERNAL] Manual Quote Request | IAF SAC - NBO Quote ID 083264310
CA-0326-02259	Re: [2nd REMINDER] SHIPMENT TO TTM/US-BT2231 @ LAX - HAWB # 6183568600
CA-0326-02625	Re: [EXTERNAL] Quote (minds)
CA-0326-02538	Re: [EXTERNAL] UPS - Shipping Quote from Richmond, VA to Los Angeles, CA : 3 DAY FREIGHT NGS
CA-0326-02359	RE: RFQ - Air freight shipment TSS, Visalia, CA to TOMRA Peru S.A.C, Peru - CIP Lima Airport - Peru - 443979 - ODQ# 085150363
CA-0326-02249	[EXTERNAL] RE: Request for Quote - Shipment to Santiago Airport > Consignee Concolor S.A.

Figure 17: LAX email volume data

Time Study Data

Process time, which is defined by the team as the average time (in minutes) needed to complete each task type, is derived from time studies that were conducted throughout the project. These values are used as fixed inputs in the calculator and are locked to prevent unintended changes. To calculate this rate, for each process, the team took the total time spent performing each task per shipment (demand received by the supercenter). Each agent is then assigned an efficiency number for each task that they performed, and the overall efficiency of that particular task was averaged across all participating agents in order to find a finalized efficiency for that task. This number was then used in the calculator as another input for the staffing capacity calculation.

Shift Duration and Utilization

The assumed idle time, excluding time for non-value-added tasks, is currently 15% based on existing UPS desk operation assumptions. For tasks such as printing documents and communications that do not contribute directly to a process bucket, users are expected to add additional idle-time allocations increasing the idle factor from the existing 15%. The process time metric represents the average time, in minutes, it takes to complete a task belonging to the process bucket. This is a constant parameter in the calculator. Hours worked per day, per agent is 8 hours or 480 minutes for every process bucket. The incoming workload is an input from the user based on existing demand/incoming email forecasts.

6.3 Calculator Structure

Structure and Application of the Calculator using

LAX is a division within the Atlanta supercenter, and the team chose to develop the initial version of the calculator for this division since all the required data was readily available. The staffing calculator Excel file is currently organized into two tabs: Inputs tab (Figure 18) and Calculation Tab (Figure 19).

STAFFING CALCULATOR — INPUTS			
DEMAND INPUTS			
Input	IAF	NAAF	Total (IAF + NAAF)
Weekly Avg OB Volume (shipments)	178	186	364
Average OB Volume from given 3 months data at LAX Supercenter	364		
PRODUCTIVITY INPUTS			
Process	Productivity (min)	Source	
Quote Request	6.41	Time Study	
Pickup Request	5.97	Time Study	
Data Entry	9.00	Time Study	
SHIFT ASSUMPTIONS			
Parameter	Value	Unit	Notes
Shift Duration	8.0	hours/day	
Days Worked per Week	5.0	days/week	
Agent Hours per Week	40.0	hrs/week	shift duration*days worked per week
Utilization Rate	85%	%	Accounts for idle/variable time
Productive Hrs/Week	34.0	hrs/week	40*0.85
QUOTE REQUEST			
Parameter	Value	Unit	Notes
Weekly Quote Demand	720	quotes/week	

Figure 18: Inputs Tab

The inputs tab contains the demand parameters, productivity and shift assumptions for each process category. The first row contains the average outbound volume which is used as the demand for pickup request and data entry process categories. Quote request demand is sourced from the LAX email volume data, which is assumed to be directly proportional to quote activity. The Productivity Inputs section contains the productivity rates for the three process categories: quote requests, pickup requests, data entry, and shipment tracking. These values, expressed in minutes per occurrence, are calculated from the time study observations conducted at the Atlanta supercenter and are locked to prevent unintentional modification. They represent the average time an agent requires to complete one instance of each task type under normal operating conditions. The Shift Assumptions section includes the number of hours worked per shift, number of days worked in a week and utilization rate. A utilization rate of 0.85 is used to account for 15% idle time.

STAFFING CALCULATIONS — ATL SUPERCENTER - LAX Division				
Parameter	Quote Request	Pickup Request	Data Entry	Notes
Weekly Demand (shipments or tasks)	720	364	364	
Productivity (min/task)	6.41	5.97	9.00	From time studies
Shift Duration	8.0	8.0	8.0	
Days Worked per Week	5.0	5.0	5.0	
Utilization Rate (Accounting for idle time)	0.85	0.85	0.85	
				Total
AGENTS NEEDED	2.26	1.06	1.61	5
Staffing Cost per Week	$ 2,624	$ 1,235	$ 1,862	$ 5,722

COLOR LEGEND	
Blue text	Enter your value
Green text	Pulled from another sheet (do not edit)
Black text	Calculated automatically

Figure 19: Calculations Tab

The calculations tab applies the staffing formula to each of the three process categories using the data from the inputs tab. For each process, the number of agents required is calculated by dividing the total workload (minutes), as the product of demand volume and productivity rate, by the available productive minutes per agent per month, which is itself derived from shift duration, days worked, and the combined unavailability factor. The result is passed through a ceiling function so that the output is always rounded up to the nearest whole agent.

The three process-level agent counts are summed to produce a total recommended headcount for the LAX division. This total represents the number of agents needed to process the forecasted shipment volume for that month across all three task categories. The staffing cost was also calculated by multiplying the number of agents with the average agents cost of $29/hour.

Extending the Model to Atlanta, Dallas, and Detroit Supercenters

With the LAX calculator validated, the same structural approach was extended to generate staffing estimates for the Atlanta, Dallas, and Detroit supercenters. The formula logic and shift assumptions in the inputs tab remain identical across all divisions. What changed between locations is the demand input and the productivity metric per supercenter. For pickup request and data entry demand, outbound shipment volume data was already available for all three supercenters and was substituted directly into the demand input fields. These figures were drawn from the same SPH spreadsheet as the average number of shipments processed for each supercenter.

Quote request demand presented a more significant challenge. The email volume data used for LAX was not available for Atlanta, Dallas, or Detroit. To address this, the team used the ratio between average email inquiries generated and average outbound shipments for the LAX division. The estimated emails was calculated using the following formula:

$$Estimated\ emails = \frac{LAX\_emails}{LAX\_Outnound\ Shipments} X\ division\_Outbound\ Shipments$$

In the above formula, LAX_emails represents the number of emails received by the LAX division in our timespan, LAX_Outbound Shipments represents the total number of outbound shipments completed by the LAX division, and division_Outbound Shipments represents the total number of outbound shipments completed by the desired supercenter. With all three demand inputs populated for each location, the staffing formula was applied to yield a recommended monthly headcount for Atlanta, Dallas, and Detroit respectively.

7. Economic Analysis

7.1 Baseline Labor Costs

An initial economic analysis was conducted by the team to determine the current baseline operating cost of the UPS Global Freight Forwarding supercenters. This baseline analysis helps determine the impact of and improvements caused by our project. From September through January, agents at the four supercenters logged a total of 130,860 labor hours. At an average wage of $29 per hour, total labor expenditures reached approximately $3.79 million, which is equal to an average of **$189,747 per week ($47,436 per supercenter).** This number includes **all agents working in the supercenters,** whether they are working on Pickup/Quote Requests/Data Entries, or other supercenter tasks. Figure 20 shows a pivot table that sorts the supercenters by metrics such as hours worked and monthly spending.

Supercenter	Year	Month	Volume of orders	Hours worked	Monthly Spending
DF1	2025	Sep	19609	5207	$ 150,997
		Oct	20002	5206	$ 150,966
		Nov	23025	6020	$ 174,590
		Dec	18987	4201	$ 121,815
	2026	Jan	18317	3810	$ 110,476
DF1 Total			99940	24443	$ 708,844
JF1	2025	Sep	35346	9142	$ 265,128
		Oct	43299	12929	$ 374,951
		Nov	51254	15869	$ 460,208
		Dec	43544	10595	$ 307,242
	2026	Jan	43478	10490	$ 304,209
JF1 Total			216921	59025	$ 1,711,737
OR1	2025	Sep	33075	4988	$ 144,646
		Oct	37150	7370	$ 213,729
		Nov	43574	7841	$ 227,399
		Dec	35447	5498	$ 159,430
	2026	Jan	34714	5110	$ 148,184
OR1 Total			183960	30807	$ 893,389
SF1	2025	Sep	14941	4339	$ 125,821
		Oct	14899	3058	$ 88,681
		Nov	17502	3392	$ 98,370
		Dec	14325	2835	$ 82,202
	2026	Jan	13305	2962	$ 85,895
SF1 Total			74972	16585	$ 480,971
Grand Total			575793	130860	$ 3,794,940

Figure 20: Baseline Spending on Staffing

Alternatively, the team has the current agent headcount for 3 of the supercenters who are supposedly assigned to Pickup/Quote Requests/Data Entries, which is shown in Figure 21.

Supercenter	Number of agents
ATL	21
DTW	20
DFW	13
Total	54

Figure 21: Current Agent Headcount for 3 Supercenters

Using the hourly wage rate of $29 per hour and 40 hours worked per week, the total number of agents calculated from Figure 21 would result in **$62,640 in weekly total labor costs ($20,880 per supercenter)**. For the sake of the economic analysis, the team will be using this number as the current labor cost for the project, as the team is observing primarily for the Pickup/Quote Requests/Data Entries buckets.

However, this information does not give the team any additional metrics regarding where agents are allocated, whether agents are full or part time, if agents are still in training, overtime worked and other important information. The current agent headcount is therefore not entirely representative of the processes that the team has analyzed. Thus, this information should only be used as a vague guideline when estimating the current baseline labor costs.

7.2 Productivity Metrics

To understand the efficiency of the supercenter operations, the company uses a productivity metric called shipments per hour (SPH). This is essentially calculated by dividing the number of processed shipments by the total labor hours. This key metric provides an estimate of how many orders are being processed per labor hour. Using the total shipment volume of 575,793 and total labor hours of 130,860 hours, the average productivity across all four supercenters is around 4.4 shipments per hour. This is a baseline value of the current operations and since these supercenters are newly established, the team is tasked with analyzing inefficiencies and idle times to improve productivity.

The average values are high level estimates of productivity and do not account for conditions like supercenter size, demand variability and agent training. So, the productivity trends were analyzed on a weekly basis across the four supercenters.

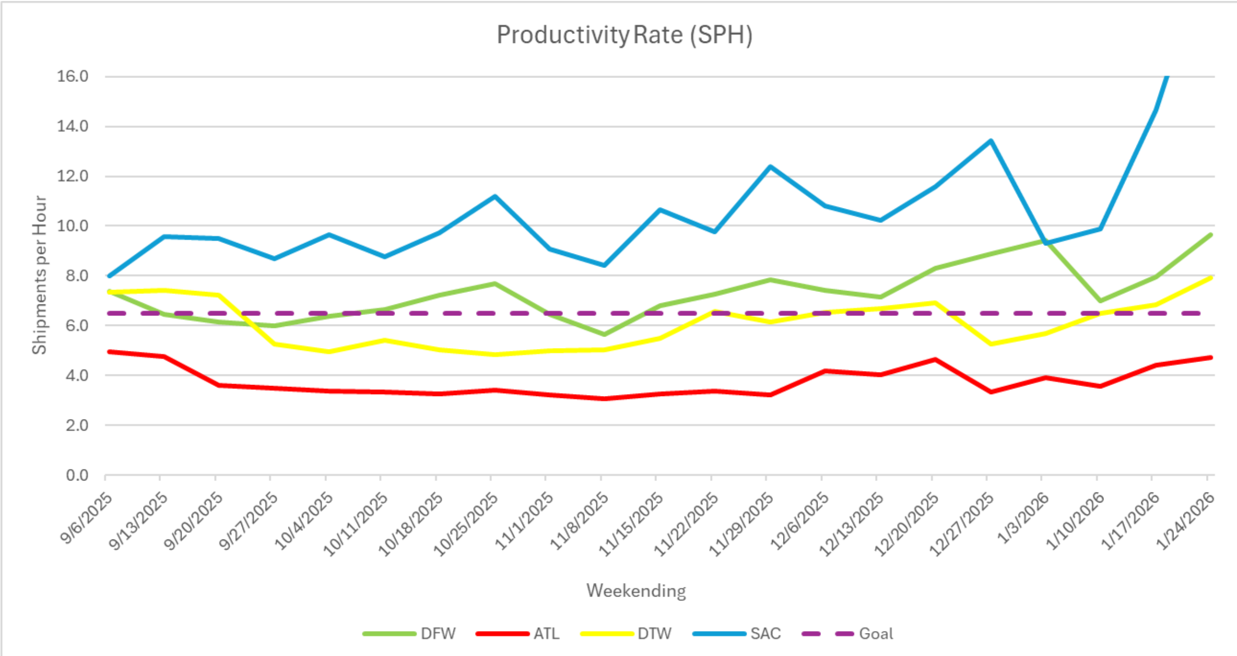

Figure 22: Weekly shipments-per-hour trends across the supercenters

Figure 22 shows differences in productivity across the supercenters. For example, the DF1 supercenter does not observe too much variability while also maintaining a high productivity level. These variations could be due to differences in shipment demand, workload distributions and task assignments. This can result in inefficient labor utilization and increased operational costs. Standardizing process times through formal time studies will enable us to identify and address the underlying reasons for these variations. Recently the UPS team has established a goal for productivity to be 6.5 shipments per hour. This is a value that the supercenters are working towards achieving. Although this number does not have math supporting it, it is the UIUC team's task to validate this or propose an alternate goal.

7.3 Volume Trends

Shipment volume trends were analyzed to understand demand fluctuations across the four supercenters. Weekly shipment data says that each facility works with different demand as shown in Figure 23.

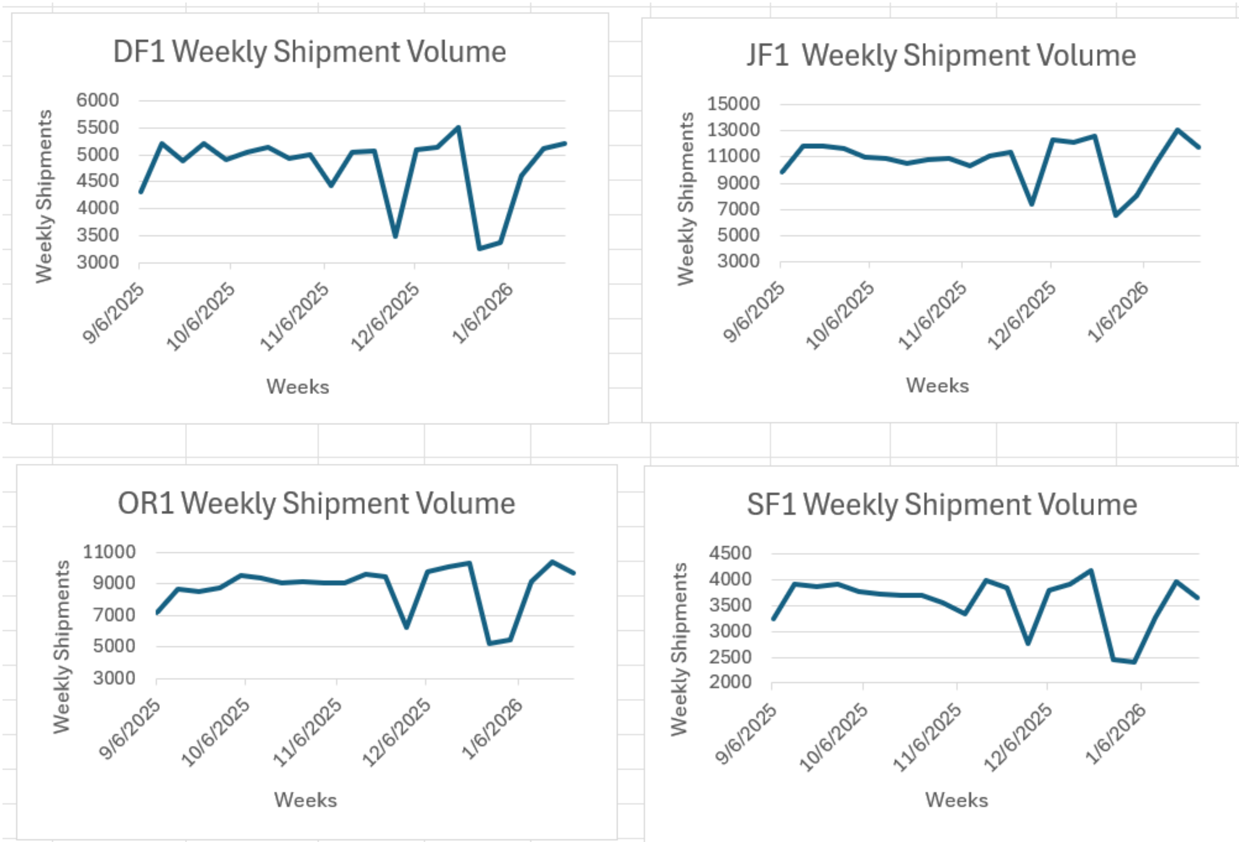

Figure 23: Weekly shipment volume trends for the four supercenters

Figure 16 shows that the JF1 supercenter processes the largest shipment volume compared to the OR1 facility that processes moderate volumes and the DF1 and SF1 process smaller volumes. The graphs also indicate the demand variability which is crucial for staffing and resource allocation. This information confirms that each supercenter would require a different number of agents due to varying shipment volumes in order to maximize productivity vs labor costs. Too much labor would incur a lot of downtime, while too little labor will not be able to meet shipment processing demands.

The values in this initial economic analysis are the baseline conditions of current supercenter operations. The time studies will provide insight into task durations and workflow inefficiencies which will be incorporated into the staffing calculator. By comparing the baseline values to the optimized values, the team can provide the company with financial savings and productivity improvements.

7.4 Economic Comparison

The team calculated a provisional headcount for DTW, DFW, and ATL supercenters based on the time studies and process analysis. Based on the time studies, process analysis, and final headcount analysis for the three supercenters, the team has determined the optimal number of agents for each supercenter shown in Figure 24.

Supercenter	Number of agents
ATL	32
DTW	16
DFW	16
Total	64

Figure 24: Final Calculated Agent Headcount for 3 Supercenters

Assuming that labor costs $29/hour and agents work 40 hours/week, the new number of agents for each of the three supercenters would result in **$74,240 in weekly total labor costs ($24,747 per supercenter)**. Compared to the previously mentioned cost of **$62,640 in weekly total labor costs ($20,880 per supercenter)** for the same three supercenters, this is a 18.52% increase in labor and labor costs from before in order to meet demands. This does not include any other labor costs associated with training, physical resources, etc.

Our results show that UPS supercenters are understaffed compared to their current states, which is contrary to UPS management's assumption. The team believes that the email data being used for the demand represents an overestimate of the actual demand, which leads to more agents being required. The team also noticed at least one agent still training for the job, which causes frequent supervisor meetings and slower process times.

Nonetheless, the team suggests that this data be considered a good baseline, with the assumptions listed in the next section. As the minimum headcount calculated by our model represents the minimum amount to satisfy the demand within a strict delay tolerance, the 18.52% increase in total headcount is a baseline to improve operational reliability. The staffing model is based on real, sampled data, as opposed to the arbitrary staffing decisions that were made in the past. Furthermore, the team was able to quantify the productivity of the main desk operations for each supercenter, and these samples, while small, can be expanded upon in future projects.

8. Assumptions

There are a few disclaimers to mention. First, the previous labor cost does not distinguish between which agents are working on which tasks, rather just a cumulative count of agent hours, regardless of the task they are assigned to. This results in large error bars when estimating the costs, because not all of these hours are specifically desk operations, which is what this project is focused on. Furthermore, there are lots of variables that should be considered when taking into account the new labor cost.

The supercenter model is a relatively new model for UPS, which means that the data collected for the previous economic analysis is very limited, and should be considered with large margins for error. The time studies cover a small sample size of overall agent activities across all of the supercenters. Our studies sample 11 agents over the span of 4 weeks in February-March, and each agent specifically for a day. Thus, the data may be affected by seasonal variations in demand, and high variability within the agents due to the sample size. UPS is in the process of training and hiring new agents constantly at the time of this report. This results in massive variability in agent productivity, extraneous tasks like training tasks and assisting new agents. Hence, the productivity constants should be treated as reasonable estimates rather than precise benchmarks until a larger and more longitudinally diverse dataset is available.

Email volume for divisions other than LAX is extrapolated using the ratio of LAX email volume to LAX outbound shipments, then applied to each division's OB data. This method assumes that the relationship between inbound quote inquiries and outbound shipment volume is consistent across divisions. Differences in customer mix, regional freight characteristics, or quote acceptance rates could cause actual email volumes to deviate from these estimates.

Furthermore, these time studies were performed virtually, which prevented the team from observing off screen activities, such as having in person meetings, conversations, and other activities that the team concluded as idle activities.

Nevertheless, the time studies and process analysis provide a good guideline for what the headcount should look like, and what the labor costs will be for that headcount.

9. Conclusions and Recommendations

Overall, the team discovered through time studies, process analysis, and the final calculator model that 2 of the 3 supercenters were understaffed and required more agents than they previously had. This calculation is done through a data-driven approach across three supercenters. Even though these conclusions come with a list of assumptions that were discussed in Section 8 of the report, the team still believes that the calculation serves as a solid foundation for any future work or processes that UPS would like to pursue in order to get a more accurate staffing model for their supercenters. The team was able to quantify and categorize the main activities performed by agents at each of the observed supercenters, and through a small sample size, develop a staffing model to obtain a quantifiable staffing result for each of the three supercenters.

Based on the time studies, process analysis, and staffing calculator developed throughout this project, the team offers the following recommendations for UPS Global Freight Forwarding to improve supercenter efficiency and optimize labor allocation across all four locations.

The primary recommendation is the monthly adoption of the dynamic staffing calculator. Because shipment demand fluctuates across weeks and months, a static headcount cannot reliably match workload to labor capacity. The calculator should be run at the start of each month using the prior month's outbound shipment volume and projected email demand as inputs. This practice will allow operations managers to make proactive staffing adjustments rather than reactive ones, reducing both the risk of overstaffing during slower periods and the service degradation that accompanies understaffing during peak demand.

Standardizing agent workflows across all supercenter locations is another critical recommendation. The time studies revealed meaningful variation in how agents at Dallas, Detroit, and Atlanta execute nominally identical tasks, with Atlanta recording the longest average cycle times across all four process categories and Dallas absorbing disproportionate communication and idle overhead. Without a documented standard operating procedure enforced consistently across locations, these discrepancies will persist regardless of calculator adoption. The team recommends that UPS develop and distribute a formal process guide derived from the standardized process maps produced in this project, with particular attention to the quote request, pickup request, and shipment entry workflows.

UPS should also invest in targeted process improvements identified during the time studies. Shipment entry was the most time-intensive task at every location, largely because agents must navigate E2K, Teams, Outlook, and ImageIndexUI sequentially for each shipment. Where customers can provide shipment data in CSV format rather than PDF or email, entry time was observed to decrease by approximately half. UPS should actively encourage this data format with high-volume customers and explore whether partial integration between these systems is feasible, as even modest reductions in entry time per shipment would compound significantly at scale.

Finally, the team recommends that UPS continue collecting structured time study data on a quarterly basis. The current study sampled 11 agents over four weeks in February and March, a window that may not capture seasonal demand variation or the productivity trajectory of agents currently in training. Refreshing the productivity constants in the calculator annually as more data becomes available will ensure the tool remains accurate and useful.

10. References

[1] United Parcel Service. (n.d.). PRQ process flow (Document No. uscaop1100) [Internal flow chart]. UPS Internal Portal.

[2] United Parcel Service. (2026). GFF Shipments by Tariff Point [Data set]. Microsoft Excel.

11. Appendix

STAFFING CALCULATIONS — ATL SUPERCENTER				
Parameter	Quote Request	Pickup Request	Data Entry	Notes
Weekly Demand (shipments or tasks)	4,190	2,361	2,361	
Productivity (min/task)	6.41	5.97	9.00	From time studies
Shift Duration	8.0	8.0	8.0	
Days Worked per Week	5.0	5.0	5.0	
Utilization Rate (Accounting for idle time)	0.80	0.80	0.80	
				Total
AGENTS NEEDED	13.99	7.34	11.07	32
Staffing Cost per Week	$ 16,227	$ 8,516	$ 12,838	$ 37,581

Figure 25: Final Calculated Agent Headcount for Atlanta Supercenter

STAFFING CALCULATIONS — Dallas Supercenter				
Parameter	Quote Request	Pickup Request	Data Entry	Notes
Weekly Demand (shipments or tasks)	2,034	1,146	1,146	
Productivity (min/task)	6.41	5.97	9.00	From time studies
Shift Duration	8.0	8.0	8.0	
Days Worked per Week	5.0	5.0	5.0	
Utilization Rate (Accounting for idle time)	0.80	0.80	0.80	
				Total
AGENTS NEEDED	6.79	3.56	5.37	16
Staffing Cost per Week	$ 7,876	$ 4,133	$ 6,231	$ 18,241

Figure 26: Final Calculated Agent Headcount for Dallas Supercenter

STAFFING CALCULATIONS — Detroit Supercenter				
Parameter	**Quote Request**	**Pickup Request**	**Data Entry**	**Notes**
Weekly Demand (shipments or tasks)	2,018	1,137	1,137	
Productivity (min/task)	6.41	5.97	9.00	From time studies
Shift Duration	8.0	8.0	8.0	
Days Worked per Week	5.0	5.0	5.0	
Utilization Rate (Accounting for idle time)	0.80	0.80	0.80	
				Total
AGENTS NEEDED	**6.74**	**3.54**	**5.33**	**16**
Staffing Cost per Week	$ 7,815	$ 4,101	$ 6,182	$ 18,098

Figure 27: Final Calculated Agent Headcount for Detroit Supercenter

Start Time	End Time	Total Time	System Used	Notes
9:02			E2K	PRQ Rate Request
9:04	9:05		Outlook email	Email with Pick up request information
9:09	9:10		Phone Call?	Phone call (went to the wrong department)Accidental
9:11	9:12		MS Teams	Responding to Aaron
9:12	9:14		E2K - PRQ	Starting Pick up request & entering information
9:14	9:15		outlook email	
9:15	9:17		phone call regarding return	confirming pickup availability
9:17	9:22		email outlook +E2K+PDK on browser	sending bill of landing + fill out the details
9:22	9:24		drafting email	
9:24	9:25		deleting emails / filler responses	
9:25	9:25		E2K - PRQ	Pickup Request Scheduled
9:25	9:27		E2K - PRQ	Pickup Request Scheduled
9:27	9:29		E2K - PRQ + outlook	Fixing pickup request (altered dates) - rescheduled
9:29	9:34		E2K - Rate Request + outlook + excel	email for rate request, filled e2k with excel
9:34	9:34		teams - discussing & planning lunch with co-worker	sushi poki - fried icecream lunch
9:34	9:38		E2K - Rate Request + outlook + word	e2k filled word doc for rate request
9:38	9:39		teams aaron - work related	
9:39	9:40		phone call regarding return - pick n hold	tracking a package - customer call,
9:40	9:42		teams	reachig out to delivery - solving customer
9:42	9:42		teams	discussing & planning lunch with co-worker
9:42	9:43		teams	talking to co-workers
9:43	9:45		outlook email	scrolling through emails
9:45	9:46		phone call regarding return - hang up	no updates found on order
9:46	9:48		E2K - CSV - customer service caller menu	from bill of landing
9:48			outlook email	scrolling

Alante Time Study | Gupreet Time Study | Barbara Time Study | Vince Time Study | Darius Time Study | Rhonda Time Study | Melissa Time Study

Figure 28: Final Time Study Spreadsheet